

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

By - Vishak Kashyap K, UG4 CND

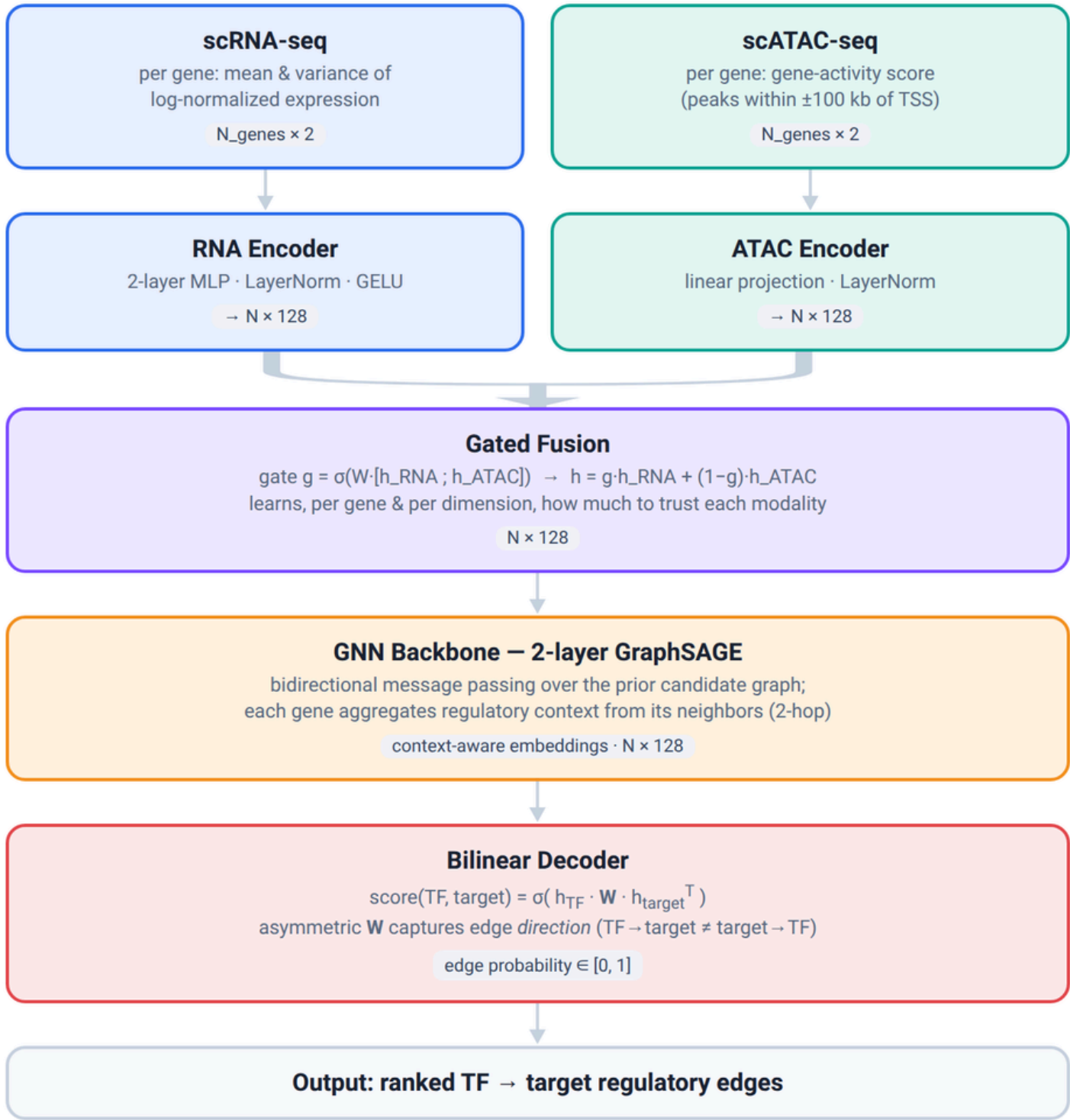
Advisor - Dr Vinod PK

10th July '26 - 21st July '26

Weekly Update

MEvD-GRN v1 Architecture

MODEL ARCHITECTURE (PER-GENE DATA FLOW)



TRAINING STRATEGY — MULTI-EVIDENCE CURRICULUM

Supervised link prediction. The database's three evidence tiers form a natural **confidence hierarchy**; we train **easy** $\rightarrow$ **hard**, fine-tuning through each tier.

Stage 1 · Localization

ChIP-seq binding — many edges, noisy signal

EASY

↓ fine-tune, lower learning rate

Stage 2 · Perturbation

knockout / CRISPR — functional & causal

HARDER

↓ fine-tune, freeze encoders

Stage 3 · Dual-Evidence

binding **AND** functional — highest confidence

HARDEST

Why a curriculum? The highest-value tier (dual-evidence) is far too small to train from scratch. Earlier, larger tiers give a strong **warm start**, so the final stage only has to *specialize*. No existing GRN method uses these three tiers as a training curriculum — this is our core contribution.

Training details. AdamW + cosine-annealing LR · class-weighted BCE (focal option) · edge-level train/val/test split (no leakage) · hard negatives mined from adjacent tiers · early stopping on validation AUPR.

Evaluation. Score every candidate edge on a held-out tier, then rank. **AUPR** (primary — networks are sparse) · **AUROC** · Early Precision · EPR (enrichment over random).

Flaws in MEdD GRN V1

MEdD-GRN infers Gene Regulatory Networks (GRNs) by integrating single-cell RNA sequencing (scRNA-seq) and single-cell ATAC sequencing (scATAC-seq) data.

- **Flaw 1 - Biologically Incoherent Early Fusion:**

- The V1 model used convex gated fusion, $(g \cdot h_{\text{RNA}}) + (1 - g) \cdot (h_{\text{ATAC}})$, forcing gene expression and chromatin accessibility to compete as interchangeable substitutes rather than complementary preconditions.

- **Flaw 2 - Loss of Co-Expression Dynamics:**

- The V1 preprocessing pipeline reduced expression data down to global mean and variance, completely discarding single-cell TF-target co-variation across cells.

- **Flaw 3 - Generic Graph Topology:**

- The V1 prior graph globally ranked accessible genes once and connected every TF to the exact same target set, carrying zero TF-specific candidate regulation.

Datasets & Feature Engineering Improvements

- **Evaluated Datasets:** Validated and retrained on benchmark human K562 (22,883 genes, 209 TFs) and mouse embryonic stem cell (ESC; 14,608 genes, 72 TFs) multi-omic datasets.
- **Expanded RNA Features:** Enhanced node features from 2 to 3 dimensions: mean, variance, detection_rate.
- **Truncated SVD Co-Expression Signatures:** Extracted 50-dimensional co-expression signatures via truncated SVD on column-standardized expression matrices.
- **Correlation Recovery:** The signature dot product directly approximates Pearson correlation with $r = 0.996$ on validation checks.
- **Target-Locus Openness Prior:** Introduced normalized $\log_{1p}(\text{number of peaks within TSS window})$ as a motif-free proxy for target locus accessibility and regulatability.

Dataset Size for the chosen cell lines

Metric	K562 (human)	ESC (mouse)
scRNA cells	~950	~8,750
scATAC cells	~1,100	~2,700
Transcription factors (TFs)	209	72
Target genes	~22,900	~14,600
Localization edges	~1.17 M	~76 K
Perturbation edges	~173 K	~69 K
Dual-evidence edges	~27 K	~7 K

Training & Graph Topology Redesign

- **Dual Multi-Relational Topology:** Replaced the static global prior with two biologically meaningful message-passing graphs:
 - Co-expression kNN Graph (Gene–Gene): Connects each gene to its top 20 co-expressed neighbors via signature dot products to capture co-regulated modules.
 - TF-Candidate Graph (TF → Target): Connects each TF to its top 500 targets that are both accessible ($\text{openness} > 0$) and co-expressed with that specific TF.
- **Strict Label Protection:** Known reference evidence edges are completely excluded from message-passing graphs to prevent validation/test label leakage.
- **Full-Space Negative Sampling:** Positives are drawn from reference networks while negative samples are drawn randomly across the full bipartite non-positive space.

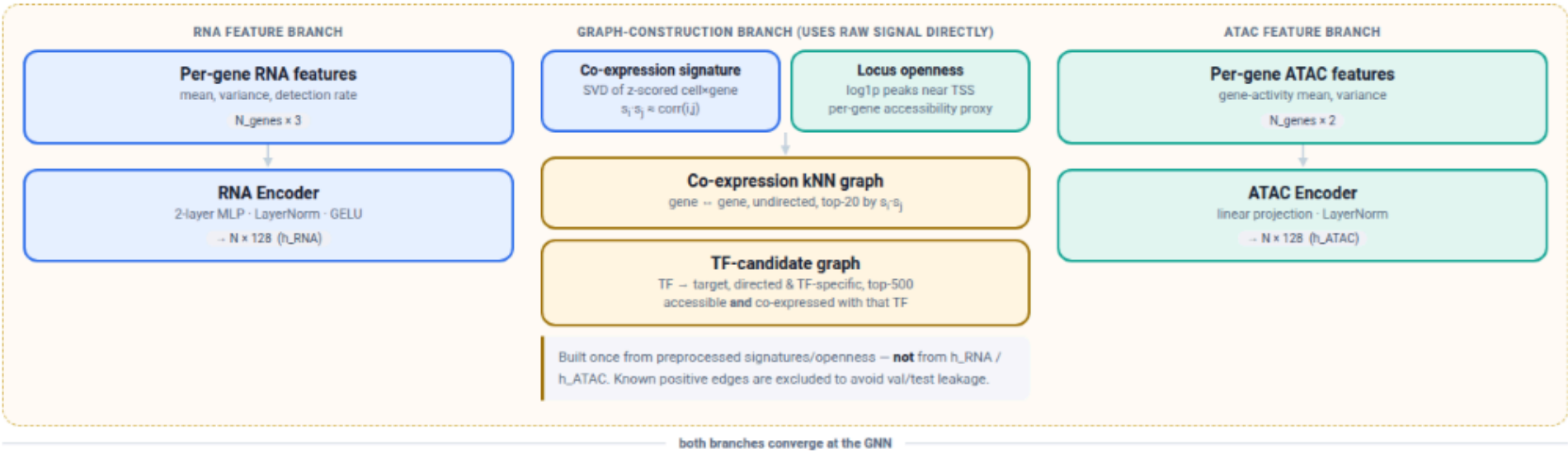
MEvD-GRN v2 Architecture

1 RAW INPUTS



2 FEATURE ENCODING & GRAPH CONSTRUCTION — RUN IN PARALLEL

Independent, simultaneous branches

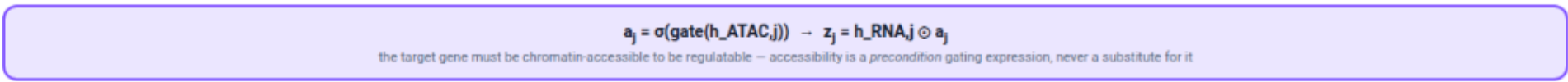


3 RELATIONAL GNN — TWO CHANNELS, TWO GRAPHS, NO EARLY FUSION



No early fusion, RNA and ATAC embeddings are propagated over the same graph structure but never averaged into one vector — each relation's SAGEConv output is summed within a channel, never across channels.

4 ACCESSIBILITY GATE (TARGET ONLY)

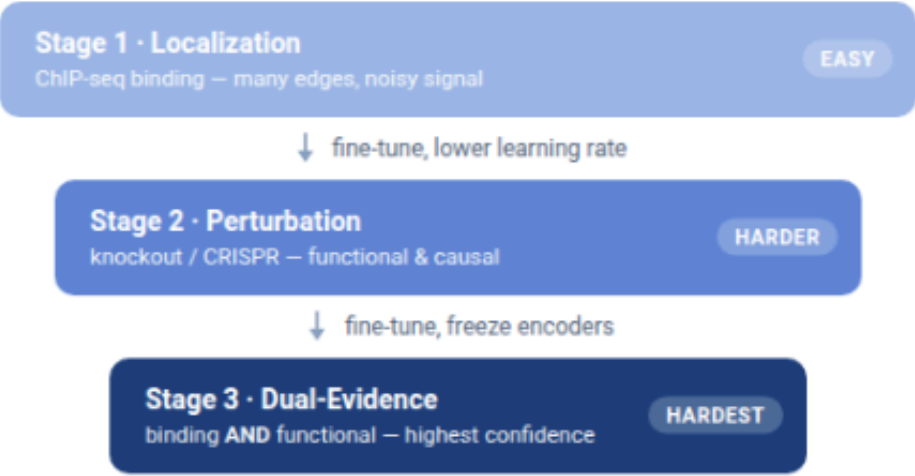


5 ROLE-AWARE EDGE DECODER (TF → TARGET)



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Training details. AdamW + cosine-annealing LR · class-weighted BCE (focal option) · edge-level train/val/test split (no leakage) · hard negatives mined from adjacent tiers · random negatives from the full bipartite space (not the message-passing graph) · early stopping on validation AUPR.

Evaluation. Score every candidate edge on a held-out tier, then rank. **AUPR** (primary — networks are sparse) · **AUROC** · Early Precision · EPR (enrichment over random).

Results

K562 — full metrics (AUPR / AUROC)

Tier	Previous (v1)	Current (v2)	GRNBoost2	GMF-GAE
Localization	0.710 / 0.639	0.712 / 0.645	0.540 / 0.506	0.541 / 0.526
Perturbation	0.337 / 0.706	0.346 / 0.678	0.184 / 0.549	0.196 / 0.562
Dual-evidence	0.878 / 0.976	0.907 / 0.984	0.175 / 0.532	0.167 / 0.530

ESC — full metrics (AUPR / AUROC)

Tier	Previous (v1)	Current (v2)	GRNBoost2	GMF-GAE
Localization	0.445 / 0.799	0.434 / 0.792	0.183 / 0.570	0.186 / 0.553
Perturbation	0.358 / 0.666	0.390 / 0.745	0.152 / 0.422	0.174 / 0.478
Dual-evidence	0.728 / 0.938	0.761 / 0.955	0.171 / 0.529	0.177 / 0.523



THANK YOU!

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